

Project Title Netraksh AI: Secure Edge Authentication for NHAI Datalake 3.0

Team Name Team Netraksh AI

Team Members

1. Aryaman M Singha — AI & Machine Learning Lead
 2. Anjali Gupta — React Native, IOS build & AWS Integration, Data Base
 3. Avishkar — Android Development & Debugging
 4. Govinda Upadhyay — Technical Documentation & Presentation
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Problem Statement

NHAI manages 5,500+ engineers and contractors working daily across India's highway construction sites — many located in remote areas with zero network coverage. Current cloud-dependent attendance systems fail completely in these dead zones, creating two critical problems. First, authentication becomes impossible without connectivity. Second, the system becomes vulnerable to attendance fraud through geo-tagged spoofing using photos, deepfakes, or digital screens during site inspections — directly compromising project accountability and financial integrity.

Proposed Solution

Netraksh AI is an offline-first facial recognition and liveness detection module built specifically for standard mid-range mobile devices — no internet required.

The system uses an InsightFace MobileFaceNet-512 TensorFlow Lite model to generate and match facial embeddings entirely on-device. Before every authentication, a dual-layer verification process is performed. MiniFASNet anti-spoofing detects presentation attacks such as printed photographs, digital screens, and replay attempts, while ML Kit-based liveness verification issues randomized blink, smile, and head-turn challenges to confirm the presence of a real human user.

All biometric data is encrypted locally using AES-256 and stored in a SQLite buffer. The moment network connectivity is restored, records automatically sync to AWS and temporary local data is purged — ensuring zero data loss and zero data bloat.

Tech Stack

Mobile Framework — React Native with TypeScript for cross-platform iOS and Android deployment

AI and Machine Learning — TensorFlow Lite, InsightFace MobileFaceNet-512, MiniFASNet Anti-Spoofing, Google ML Kit Face Detection & Liveness

Data and Security — SQLite for offline buffering, AES-256 for biometric hash encryption

Cloud Sync — AWS API Gateway with DynamoDB

Testing — Jest automated unit testing suite

Expected Outcomes

Ultra-Low Latency — End-to-end authentication processing under 0.1 seconds, significantly outperforming the 1-second NHAH requirement.

High Accuracy — Greater than 95% verification accuracy across harsh environments including dawn, noon, and night using adaptive thresholding and anti-spoof verification.

Zero Data Bloat — Total model payload of approximately 12.25 MB, comfortably within the 20 MB deployment constraint.

Fraud Prevention — Protection against static photos, replay videos, digital screens, and presentation attacks through MiniFASNet anti-spoofing and active liveness verification.

Challenges That We Overcame

1. Model Size vs Accuracy Trade-Off

Balancing recognition accuracy, anti-spoofing capability, and model footprint within the strict 20 MB constraint was our primary challenge. By selecting InsightFace MobileFaceNet-512 (6.55 MB) and MiniFASNet (5.70 MB), we achieved a combined deployment size of approximately 12.25 MB while maintaining high recognition performance.

2. Cross-Platform Device Compatibility

Ensuring consistent TensorFlow Lite model performance across Android and iOS devices with varying camera APIs, native permissions, and hardware capabilities required extensive testing and optimization.

3. Offline Anti-Spoofing and Liveness Verification

Implementing both anti-spoof detection and active liveness verification entirely offline required integrating MiniFASNet with ML Kit-based blink, smile, and head-turn detection without relying on cloud-based security services.

4. Secure Biometric Storage at Speed

Implementing AES-256 encryption for face embeddings in SQLite while maintaining real-time retrieval performance required careful database optimization and memory management.

5. Harsh Field Condition Reliability

Building adaptive thresholding logic to maintain recognition accuracy across low-light, high-glare, and nighttime conditions specific to Indian highway construction sites required extensive image quality analysis and threshold calibration.

6. Zero Data Loss Sync Architecture

Designing a reliable offline queue capable of surviving app restarts, failed synchronizations, and intermittent connectivity while automatically purging data after successful AWS synchronization required careful state management across SQLite and network layers.

Timeline

25th May — Discovery & Planning Conducted full requirement analysis of the NHAIDatalake 3.0 hackathon problem statement. Identified technical gaps, defined project scope, assessed feasibility constraints (20MB model, sub-1s latency, offline-first), and classified team roles based on individual strengths.

26th May — Research & Architecture Decision Deep-dived into available face recognition models — FaceNet, ArcFace, MobileFaceNet — and benchmarked them against the hackathon constraints. Selected ArcFace-MobileNetV2 with TFLite quantization as the optimal solution. Finalized tech stack decisions for React Native, MediaPipe, SQLite and AWS.

27th May — System Design Developed complete system architecture including flowcharts, Data Flow Diagrams, authentication workflow, ER diagram for local SQLite schema, and offline sync and purge mechanism design.

29th May — Environment Setup Set up development environments across all team members' machines. Configured IDEs, React Native toolchains, Android Studio, Xcode, and Node version management for consistent cross-platform builds.

30th May — Dependency Integration Resolved AI model library compatibility issues across Android and iOS OS versions. Installed and configured TensorFlow Lite, MediaPipe Face Mesh, React Native Vision Camera, and op-sqlite with native bindings.

1st June — Foundation Build Completed basic React Native app structure with navigation. Set up Axios client with interceptors and error handlers for AWS API communication. Began technical documentation skeleton and defined basic model architecture. Initialized SQLite database schema with migrations.

2nd June — Core Feature Development Designed and finalized project logo. Built UI screens and camera capture functionality. Completed SQLite database setup with encrypted face template storage. Began iOS device build configuration. Integrated initial MobileFaceNet model and began ArcFace-MobileNetV2 integration.

3rd June — Integration & Cloud Completed Android device build configuration and resolved platform-specific compatibility issues. Synced AI pipeline with the React Native app. Set up AWS API Gateway and DynamoDB for cloud sync. Updated all technical documentation. Integrated live video embedding for real-time face detection. Upgraded to full ArcFace-MobileNetV2 with multi-face detection support.

4th June — Testing & Submission Prep Completed PPT presentation and project proposal. Integrated all AI models with UI for end-to-end authentication flow. Conducted black-box testing of the full authentication pipeline. Verified zero data bloat — model payload confirmed at 5.2MB. Updated benchmark documentation with real inference results — 1.35ms matching latency, 99.7% accuracy.

5th June — Final Testing & Submission Conducted final end-to-end system testing across Android and iOS devices. Verified offline authentication, liveness detection, sync and purge mechanism. Completed GitHub repository clean up and submitted all deliverables including source code, PPT and technical documentation.

Executive Summary

Netraksh AI is an offline-first mobile authentication system built for NHAI's field operations. It solves a critical real-world problem — authenticating 5,500+ engineers and contractors daily at remote highway construction sites where internet connectivity is completely unavailable.

The system performs facial recognition and liveness detection entirely on-device using a compressed ArcFace-MobileNetV2 model at just 5.2MB — well under the 20MB constraint. Before every authentication, a randomized liveness challenge confirms human presence, preventing spoofing via photos, deepfakes, or digital screens. All biometric data is encrypted using AES-256 and stored locally in SQLite. The moment connectivity is restored, records automatically sync to AWS DynamoDB and temporary local data is purged.

Netraksh AI delivers 99.7% recognition accuracy, 1.35ms matching latency, and full cross-platform support for Android and iOS — exceeding every NHAI technical requirement while maintaining complete user privacy through on-device processing.

Future Enhancements

1. Passive Liveness Detection Replace active challenge-response with passive skin micro-texture and blood flow analysis — verifying human presence without asking the user to blink or turn their head. This will dramatically speed up authentication for high-volume site check-ins.

2. Multi-Face Batch Processing Enable simultaneous scanning of multiple faces in a single camera frame — allowing entire teams to authenticate during group briefings or site transport check-ins, drastically reducing daily roll-call time.

3. Hardware Acceleration Integrate NNAPI pipelines for Android and CoreML for iOS — tapping into dedicated AI processing units on modern smartphones to achieve even faster inference on budget devices.

4. Grafana Operations Dashboard Deploy a real-time Grafana dashboard for NHAI administrators showing attendance metrics, device performance, sync status, and site-specific compliance reports across all project locations.

5. Geofence-Based Authentication Add GPS boundary validation so attendance is only recorded when the employee is physically within the defined project site perimeter — preventing remote check-in fraud entirely.

6. Voice-Based Fallback Authentication Implement offline voice recognition as a secondary authentication method for edge cases where face detection quality is compromised due to extreme environmental conditions.

7. Wearable Device Integration Extend authentication support to smartwatches and wearable badges for hands-free check-in at active construction zones where holding a phone is impractical or unsafe.

8. Federated Learning for Indian Demographics Continuously improve model accuracy for diverse Indian facial demographics using federated learning — training on anonymized edge data without ever sending biometric information to a central server.

Team Netraksh AI — Secure Offline Authentication for Bharat's Infrastructure
